Graphing Basic Polar Curves from Equations

Plotting polar points, circles and lines, limacons and cardioids, and rose curves

Directions:

- All angles θ are measured in radians.
- Give exact values where an exact value exists; otherwise round to two decimal places.
- Work in the space under each problem and write the final answer on the answer line.

1. The cardioid $r = 4 + 4\cos\theta$ is graphed one point at a time. Find the exact value of r when $\theta = \dfrac{\pi}{3}$.

Answer: ________________

2. The graph of $r = 6\cos\theta$ is a circle that sits on the polar axis. Find r when $\theta = \dfrac{\pi}{3}$, which locates one point of that circle.

Answer: ________________

3. A rose curve reaches the pole at every angle where $r = 0$. Find every θ in $[0, \pi)$ for which $r = 5\sin(3\theta)$ equals zero.

Answer: ________________

4. Consider the cardioid $r = 4 - 4\sin\theta$.

(a) Find r when $\theta = \dfrac{3\pi}{2}$. $r =$ ________

(b) Explain why this curve touches the pole at exactly one angle in a single full turn.

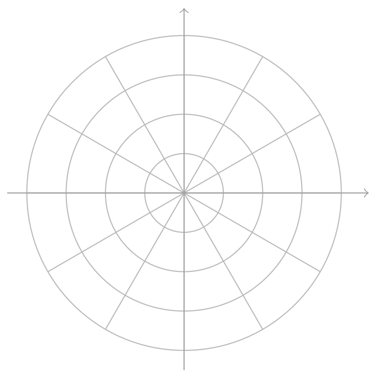
5. Find the exact value of r for the limacon $r = 2 + 3\cos\theta$ when $\theta = \dfrac{2\pi}{3}$.

Answer: ________________

6. Consider the rose $r = 3\cos(2\theta)$.

(a) Find the exact value of r when $\theta = \dfrac{\pi}{6}$. $r =$ _________

(b) Sketch the whole curve on the polar grid below. Label the tip of the petal that lies on the polar axis with its distance from the pole.



7. The circle $r = 8\sin\theta$ passes through the pole. Find every θ in $[0, 2\pi)$ for which $r = 0$.

Answer: __________________

8. Consider the limacon $r = 2 + 4\cos\theta$.

(a) Find r when $\theta = 0$. $r =$ ________

(b) Find r when $\theta = \pi$. $r =$ ________

9. For the rose $r = 6\sin(2\theta)$, find r when $\theta = \dfrac{\pi}{8}$. Round to two decimal places.

Answer: ________________

10. The limacon $r = 1 + 3\sin\theta$ has an inner loop.

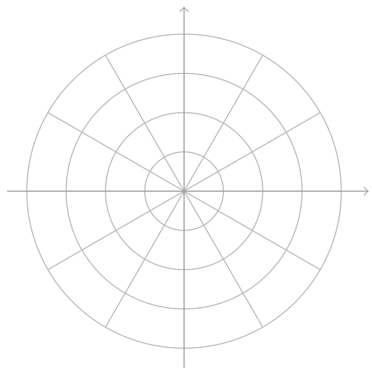
(a) Find r when $\theta = \dfrac{3\pi}{2}$. $r =$ ________

(b) Describe where that point actually lands on the plane, and say which part of the graph the negative r-values trace out.

11. The circles $r = 6\cos\theta$ and $r = 6\sin\theta$ cross each other twice.

(a) Solve $6\cos\theta - 6\sin\theta = 0$ for all θ in $[0, 2\pi)$. $\theta =$ __________

(b) Sketch both circles on the polar grid below and mark the crossing point that is not the pole.



12. A point is given in polar form as $\left(-4, \dfrac{\pi}{6}\right)$.

(a) Rewrite it with a positive first coordinate and an angle in $[0, 2\pi)$: $r =$ __________ $\theta =$ __________

(b) Explain why your rewritten pair and the original pair mark the same location.